

APOLLO

Digital Twin AI Monitoring System

OPERATIONAL

USER

MANUAL

Comprehensive

Guide for

Campus

Security

Workflows

System

Created by

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SYSTEM OVERVIEW

APOLLO Control System is a web-based CCTV monitoring system designed for BatStateU campus security and the APOLLO digital twin. The system connects to RTSP streams sources to perform real-time AI analytics. All this analytics is then sent through an API into the APOLLO digital twin.

Core Capabilities:

- YOLO-based person detection.
- Real-time count within configured Regions of Interest (ROI).
- University gate entry/exit tracking.
- Historical analytics and anomaly alerting.
- Web-based dashboard with live feeds and management tools.

GETTING STARTED

FIRST TIME SETUP

To initialize the system on a new machine, ensure the following requirements are met:

- Python 3.10 or newer.
- Network access to RTSP camera sources.
- Write permissions for the project folder.

```
.\setup_deployment.ps1
.\.venv\Scripts\python.exe deploy_check.py
.\.venv\Scripts\python.exe cctv_manager.py
```

Once started, access the dashboard at: `http://<server-ip>:5000`

USER INTERFACE & FEATURES

LOGIN

The application requires authentication. Default credentials (unless overridden by environment variables) are:

- **Username:** admin
- **Password:** Defined in config.py or via DT_WEBAPP_PASSWORD

Note: If no Password is configured, a temporary password will be generated, see on Terminal

CAMPUS DASHBOARD

The primary control center for the campus operator.

- View entry, exit, and estimated total occupants.
- Global controls to start/pause all cameras or specific gate cameras.
- Per-zone count cards with individual controls.

LIVE VIEWER

Browse and interact with real-time video feeds.

1. Select a Zone to view.
2. Click **Start Feed** for specific cameras.
3. Click a feed image to enlarge the preview.
4. Use the **Gear Icon** to access individual camera settings.

ADDING NEW STREAMS

1. Open **Add Streams** from the sidebar.
2. Enter camera name and RTSP URL (use '0' for local cameras).
3. Assign a Zone and Floor.
4. Click **Add Stream**.

ADDING NEW ZONES

1. Open Add Zones from the Sidebar
2. Enter Zone Name
3. Click Create Zone

ADVANCED CONFIGURATION

ROI (REGION OF INTEREST) SETUP

Limits detection to a specific area to reduce false counts.

1. Go to Camera Settings > **Configure ROI / Gate**.
2. Upload a CCTV reference image.
3. Click points to draw the area.
4. Click **Save ROI Setup**.

GATE COUNTER SETUP

Tracks campus entry/exit by monitoring a separator line.

1. In ROI Configuration, enable "gate counter for this camera".
2. Assign the role (Entrance or Exit).
3. Draw exactly two points to create a separator line (Always Start from left side).
4. Define crossing direction and save.

Note: Counts Reset Automatically at Midnight.

ANOMALY NOTIFICATIONS

Automated alerts for unauthorized presence during restricted times. Access the Anomaly Configuration from the Sidebar to configure which cameras will trigger anomalies and the time of restrictions.

- **Default:** Uses a global restricted time window.
- **Custom:** Set specific windows per camera.
- **Safe:** Disable anomaly tracking for specific cameras.

MAINTENANCE & OPERATIONS

HISTORICAL DATA AND REPORTING

- **Charts:** View trends for the last 24 hours on the dashboard.
- **Export:** Download CSV and graph Image reports for ranges (10h, 24h, 7d, 30d, or custom).

DATABASE MANAGEMENT

Access database stats to monitor file size. Use the **Cleanup** tool to remove old analytics and anomaly logs while preserving system configurations (ROIs, Zones, etc.).

- **Custom Range:** Keep all data from the selected range.
- **Delete All:** Delete every historical data existing on the database

TROUBLESHOOTING QUICK-LINK

Issue	Possible Cause	Checklist
Black Video Feed	Slow Internet Connection/Failed to load camera stream	Verify RTSP URL, credentials, and camera power state.
Model Load Fail	Incorrect Model path/ No model existing	Check MODEL_PATH in config.py and ensure file exists.
No Anomalies	Restricted Time or Cameras are marked Safe	Ensure camera is not marked "Safe" and count is fluctuating from zero.
Cameras Offline	Can't load camera/ slow internet connection/ Too many cameras are running	Check if the internet connection is good. Don't run too many cameras at once.

Program Suddenly Stop	Memory Allocation Issue	Verify through terminal. Restart the System.
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FILE AND SYSTEM STRUCTURE

File	Responsibility
cctv_manager.py	App Entry point & Service Initialization
routes.py	API Endpoints & Page Routing
camera_streams.py	YOLO Inference & Frame Processing
Camera_registry.py	Thread-safe in-memory camera state: counts, activity flags, groups, floors, gate totals, and config references
database.py	SQLite Storage & Schema
anomalies.py	Alert Rules & Logic
config.py	Global Constants & Tuning
Index.html / style.css	Frontend

BUGS

BUGS TO LOOK OUT FOR

1. Only the device who executed the program can view the camera but other devices can view the numbers of analytics through the Ip address of the device that executed the program.
2. Optimize the Database more to prevent bloating.

IMPROVEMENTS

THINGS TO IMPROVE

1. To avoid computational bottleneck. Each area should have separate device to run the backend so that all cameras can run simultaneously.
2. Improve AI model for better accuracy.

